

Character Representation & Data Measurement Units

Storage Units

Bit, Byte, KB, MB, GB, TB, PB

Coding Systems

BCD, ASCII, EBCDIC, Unicode

Conversions

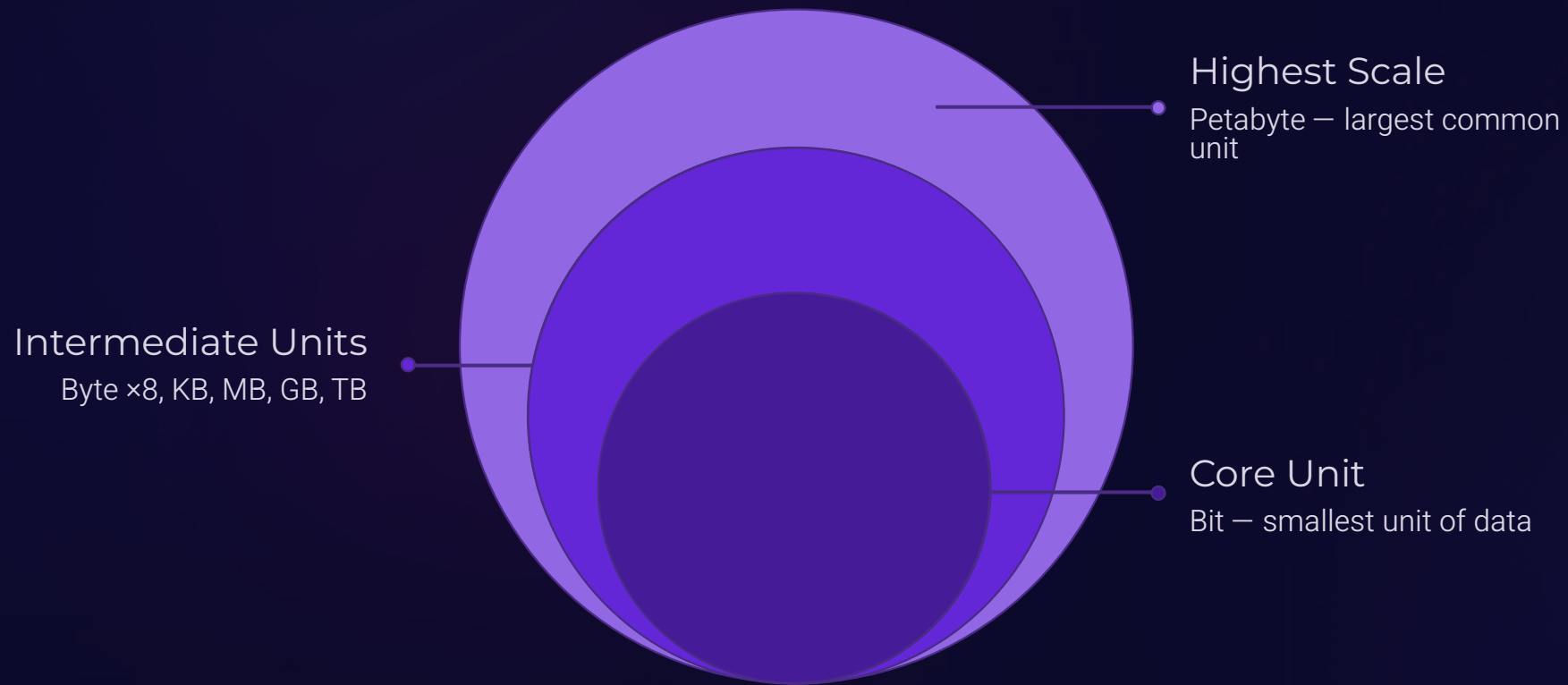
Convert between all storage units

Speed & Cost

Access speed and cost per unit

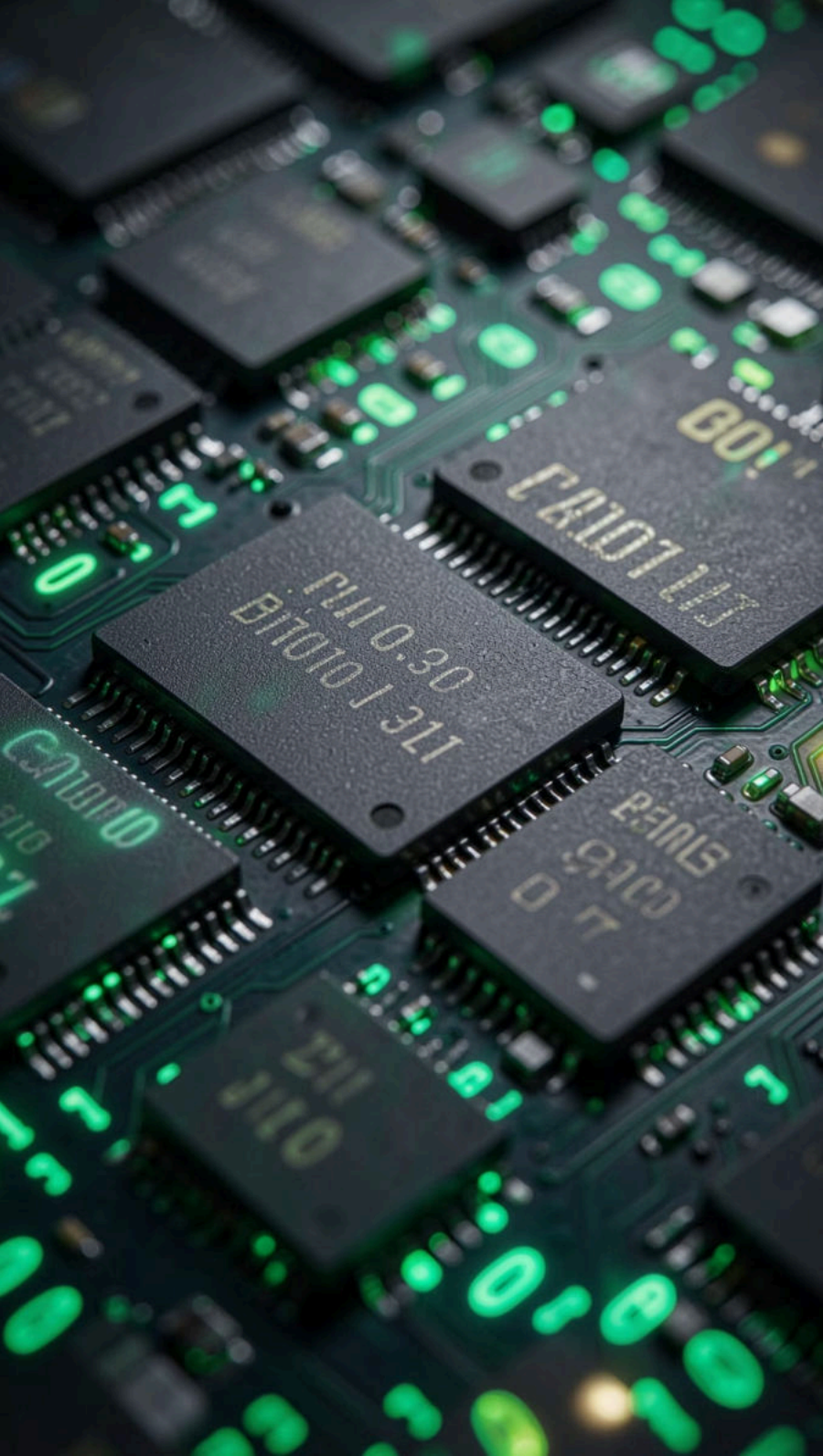
Data Storage Capacity

Every piece of data needs space. Computers measure storage using a hierarchy of units — each one **1,024 times larger** than the last. The smallest unit is the **bit**; the largest commonly used unit is the **petabyte**.



Each step up multiplies capacity by 1,024 (2^{10}), forming a clear and consistent storage hierarchy.

ICT Grade 10 - Chapter 3 - Module 3 - Akila Savinda



Storage Units — The Basics



Bit (b)

The **smallest** unit. A binary digit: **0** or **1**. Short for "Binary DigIT."



Nibble

4 bits — half a byte. Rarely used in everyday computing.



Byte (B)

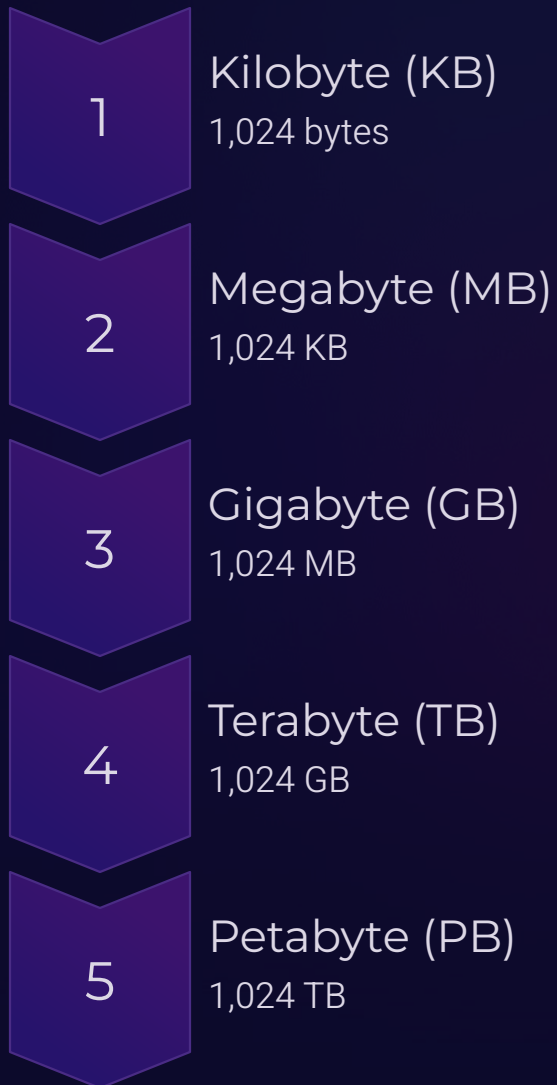
8 bits. The standard unit for storing one character.



Key relationships: 1 Byte = 8 Bits | 1 Nibble = 4 Bits = ½ Byte

Storage Units — KB to PB

Each unit is **1,024 times** the previous (2^{10}). Note: **GB** = Gigabyte; **Gb** = Gigabit — never mix them!



Real-World Storage Examples

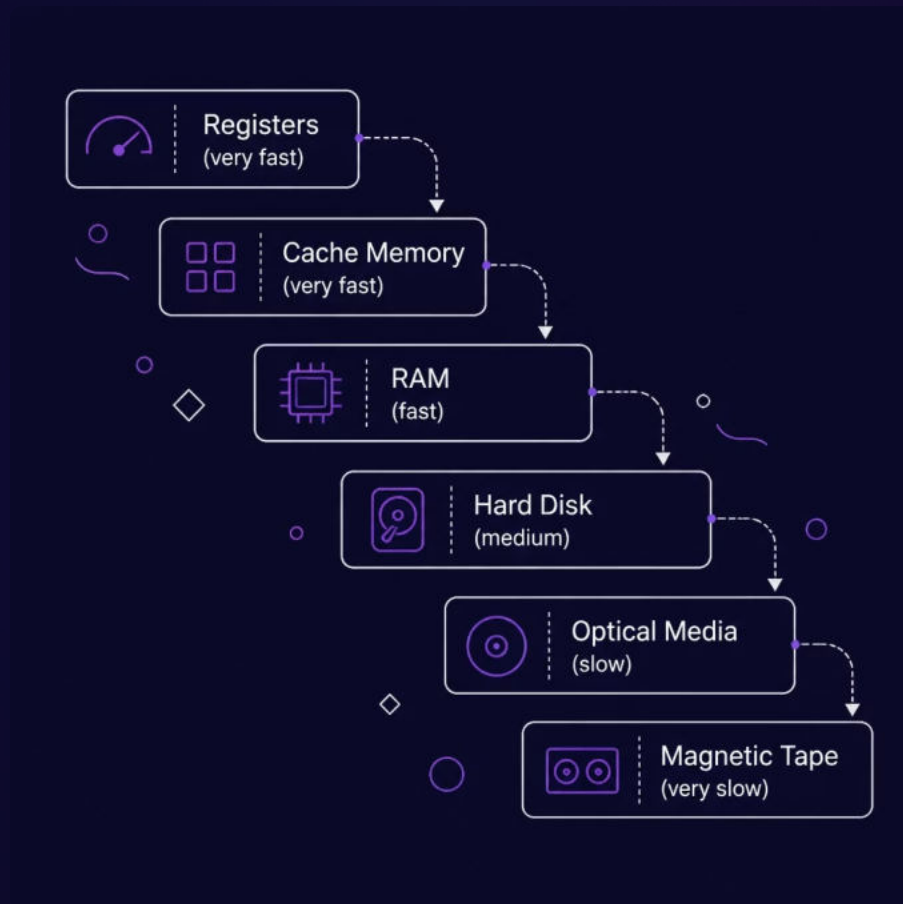
How much can each unit hold? These comparisons make the numbers meaningful:

- **1 KB** — About half a page of text
- **1 MB** — ~500 pages, 1 photo (2–5 MB), 1 song (3–5 MB)
- **1 GB** — ~500,000 pages (a large library), 1 movie (700 MB–4 GB)
- **1 TB** — ~500 million pages (a massive library)



Data Access Speed

Different storage devices take different amounts of time to read and write data. **Registers** are the fastest; **magnetic tape** is the slowest.



Key Insight

Speed and cost are directly linked. The fastest storage is the most expensive per unit. Registers and cache provide near-instant access but cost far more than hard disks or tape.

i Registers → Cache → RAM → HDD → Optical
→ Tape
Fastest → Slowest

Cost per Unit Storage

Cost per bit varies greatly by device. The fastest storage is the **most expensive**; slower media is **cheaper** per unit.

Cost Hierarchy

Most Expensive → Cheapest

Registers

Most expensive

Cache Memory

Very expensive

RAM

Expensive

Optical Media

Medium cost

Hard Disk

Cheap

Magnetic Tape

Cheapest





How Computers Represent Data

All data — text, images, sound — is converted into **binary patterns** (0s and 1s). **Coding systems** define these patterns. For example, the letter '**A**' in ASCII is **1000001**.

BCD

4 bits · 16 symbols · Early decimal numbers

ASCII

7 bits · 128 characters · English text

EBCDIC

8 bits · 256 characters · IBM mainframes

Unicode

16 bits · 65,536 symbols · All languages

Coding Systems — BCD & ASCII

BCD — Binary Coded Decimal

Uses **4 bits** per digit. Represents decimal numbers 0–9.

Example: $37_{10} \rightarrow 3=0011, 7=0111 \rightarrow \mathbf{00110111}$

 Only 10 of 16 possible codes are used.

ASCII — 7-Bit Text Encoding

ANSI-approved. Uses **7 bits** → **128 characters** (letters, digits, symbols).

Example: 'A' = 65 = **1000001**

"School" in binary:

1010011 1100011 1101000 1101111 1101111 1101100



Activity: Write the ASCII binary code for "ICT."

EBCDIC, Unicode & Module Summary

EBCDIC

8 bits → 256 characters. Used on IBM mainframes.

Unicode

16 bits → 65,536 symbols. Supports all languages (Sinhala, Tamil, Japanese, Chinese) plus emojis and special symbols. Designed by the Unicode Consortium.

Coding Systems Comparison

System	Bits	Characters
BCD	4	16
ASCII	7	128
EBCDIC	8	256
Unicode	16	65,536

❏ **Key Takeaways:** Bit → Byte (8 bits) → KB → MB → GB → TB → PB (each ×1,024). Speed: Registers → Cache → RAM → HDD → Optical → Tape (fastest to slowest). Cost: Registers (most expensive) → Tape (cheapest).